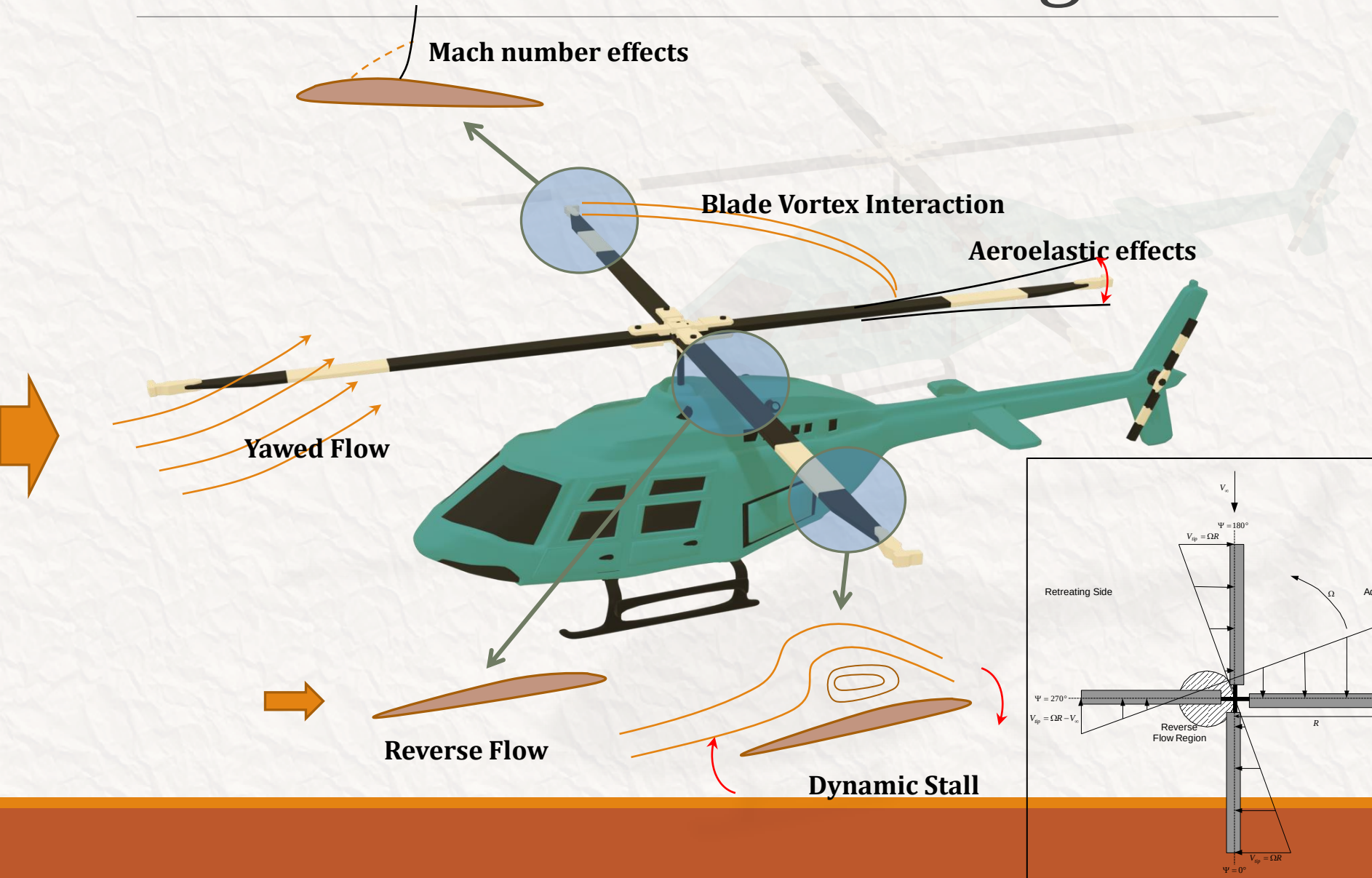
Rotary Wing Aerodynamics (AE 667)

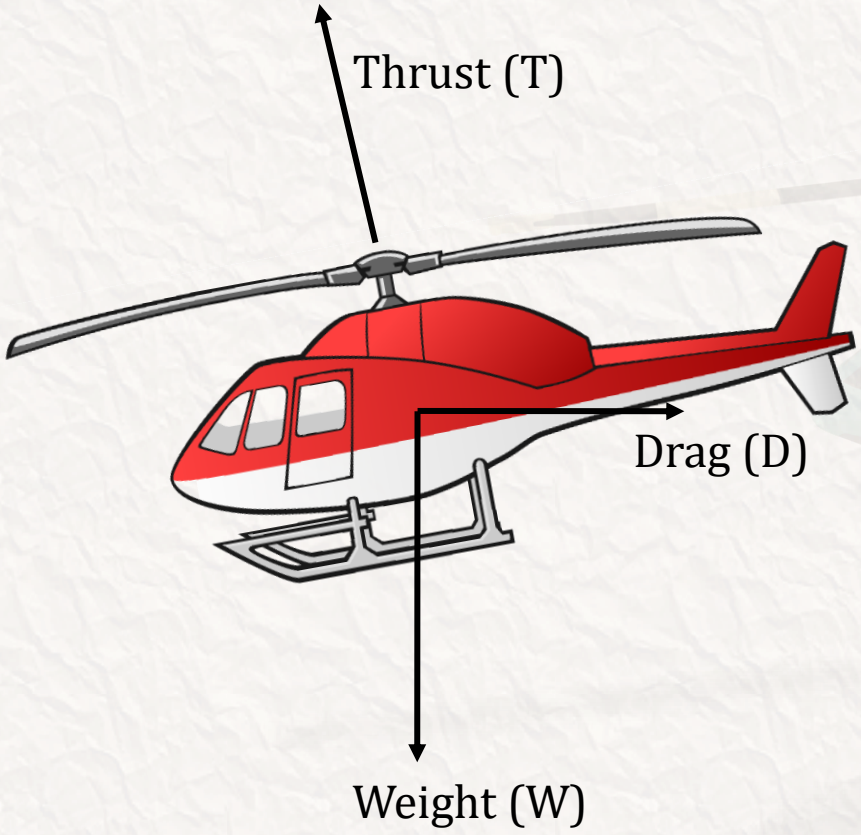


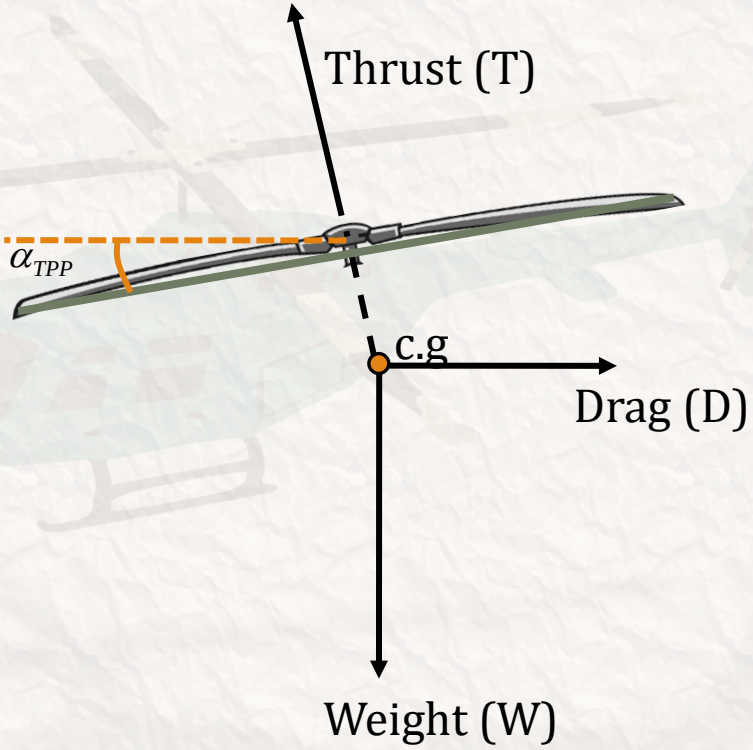
Problems of Forward Flight



Performance Prediction

Force Balance on a Helicopter



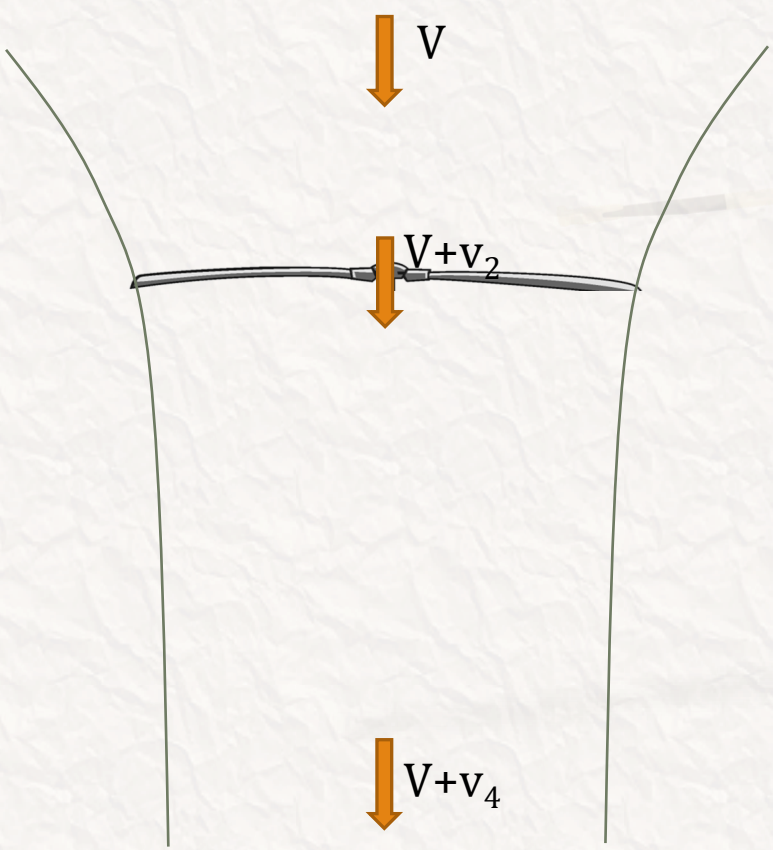


For a steady level flight :

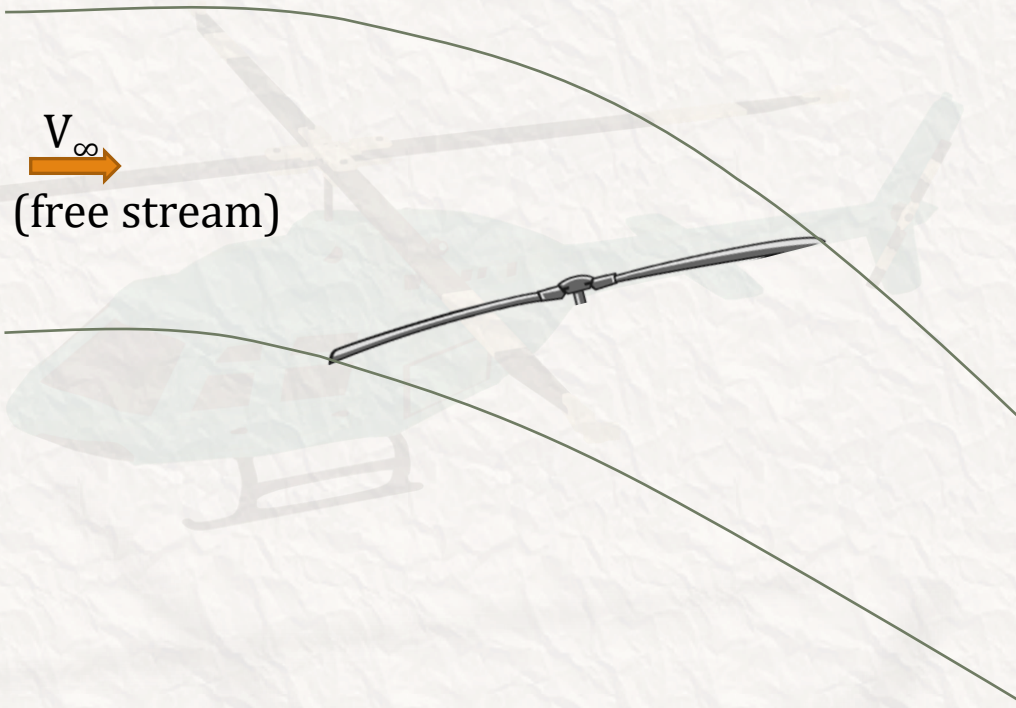
$T \sin \alpha_{TPP} = D;\ T \cos \alpha_{TPP} = W$

α_{TPP} = Tip Path Plane

Glauert's Conceptual model for Inflow



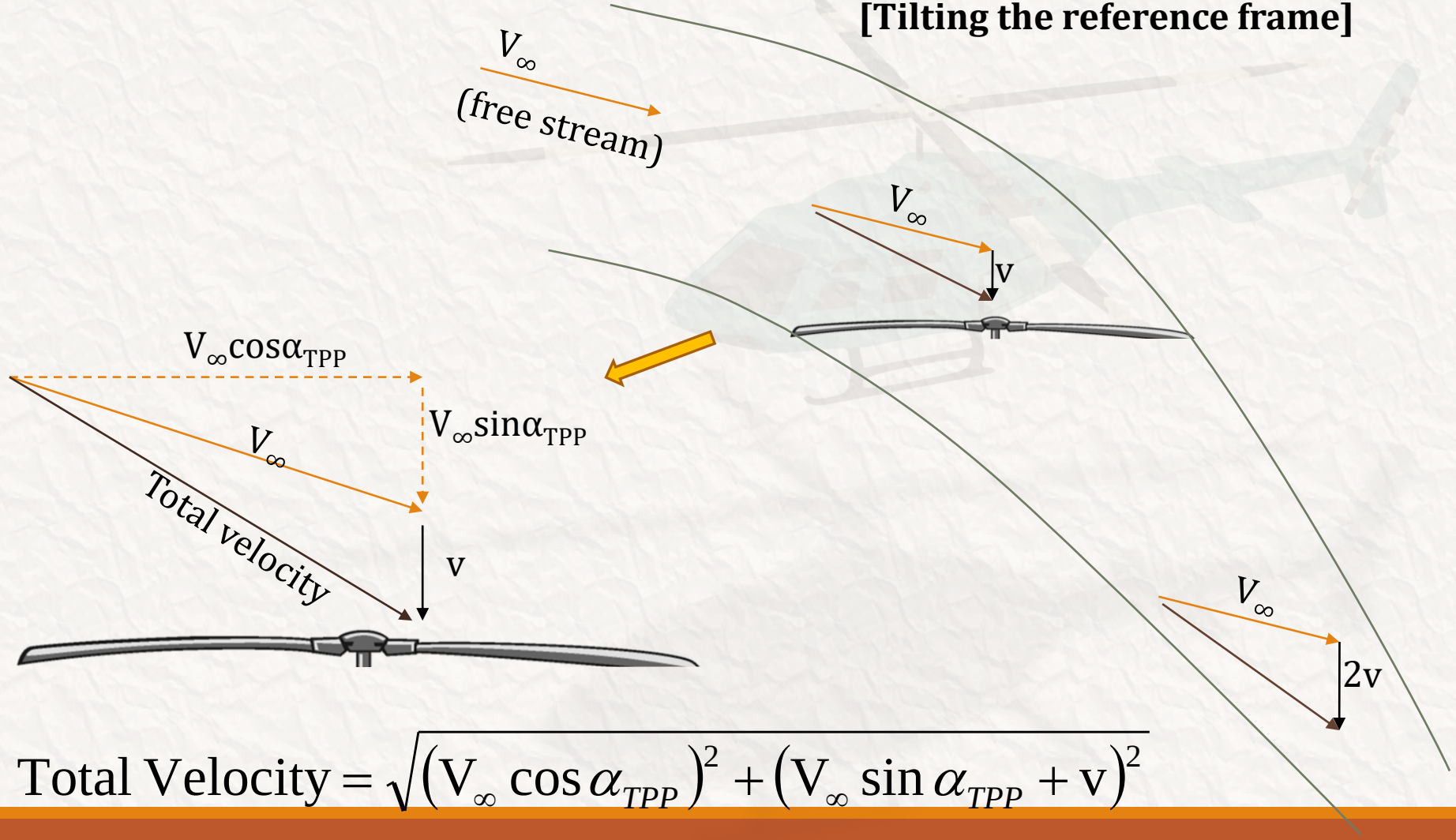
Momentum Theory for Climbing / Hovering Rotor (V is climb velocity)



Rotor in Forward Flight

Glauert's Conceptual model for Inflow

[Tilting the reference frame]



$$\text{Total Velocity} = \sqrt{\left(V_\infty \cos\alpha_{TPP}\right)^2 + \left(V_\infty \sin\alpha_{TPP} + v\right)^2}$$

Thrust estimate using Glauert's model

- In hover by momentum theory,

  Thrust = (mass flow rate)*change in induced velocity
  = ρA(V+v)(2v)

- In forward flight,

$$T = \rho A \sqrt{\left(V_\infty \cos \alpha_{TPP}\right)^2 + \left(V_\infty \sin \alpha_{TPP} + \mathrm{v}\right)^2} (2\mathrm{v})$$

Non –Dimensional Forms

- Advance Ratio:

$$\mu = \frac{\text{Edgewise Freestream Component}}{\text{Tip Speed}} = \frac{V_\infty \cos\alpha_{TPP}}{\Omega R} \approx \frac{V_\infty}{\Omega R}$$

- Coefficient of Thrust:

$$C_T = \frac{T}{\rho A(\Omega R)^2} = 2\lambda_i \sqrt{\mu^2 + \left(\mu \tan\alpha_{TPP} + \lambda_i\right)^2}$$

- For $\mu > 0.2$ and small α_{TPP}, $\lambda_i << \mu$

$$C_T \approx 2\mu\lambda_i; \quad \therefore \lambda_i \approx \frac{C_T}{2\mu}$$ (Inflow velocity decreases with advance ratio)

Power Estimate

- Induced Power = Thrust*Inflow

$$= T(V_\infty \sin\alpha_{TPP} + v)$$

$$= T \sin\alpha_{TPP} V_\infty + Tv$$

$$= DV_\infty + Tv$$

- Total power = Induced power + Blade profile power

$$= DV_\infty + Tv + \text{Profile Power}$$

Vehicle Drag Power

Hover Induced Power

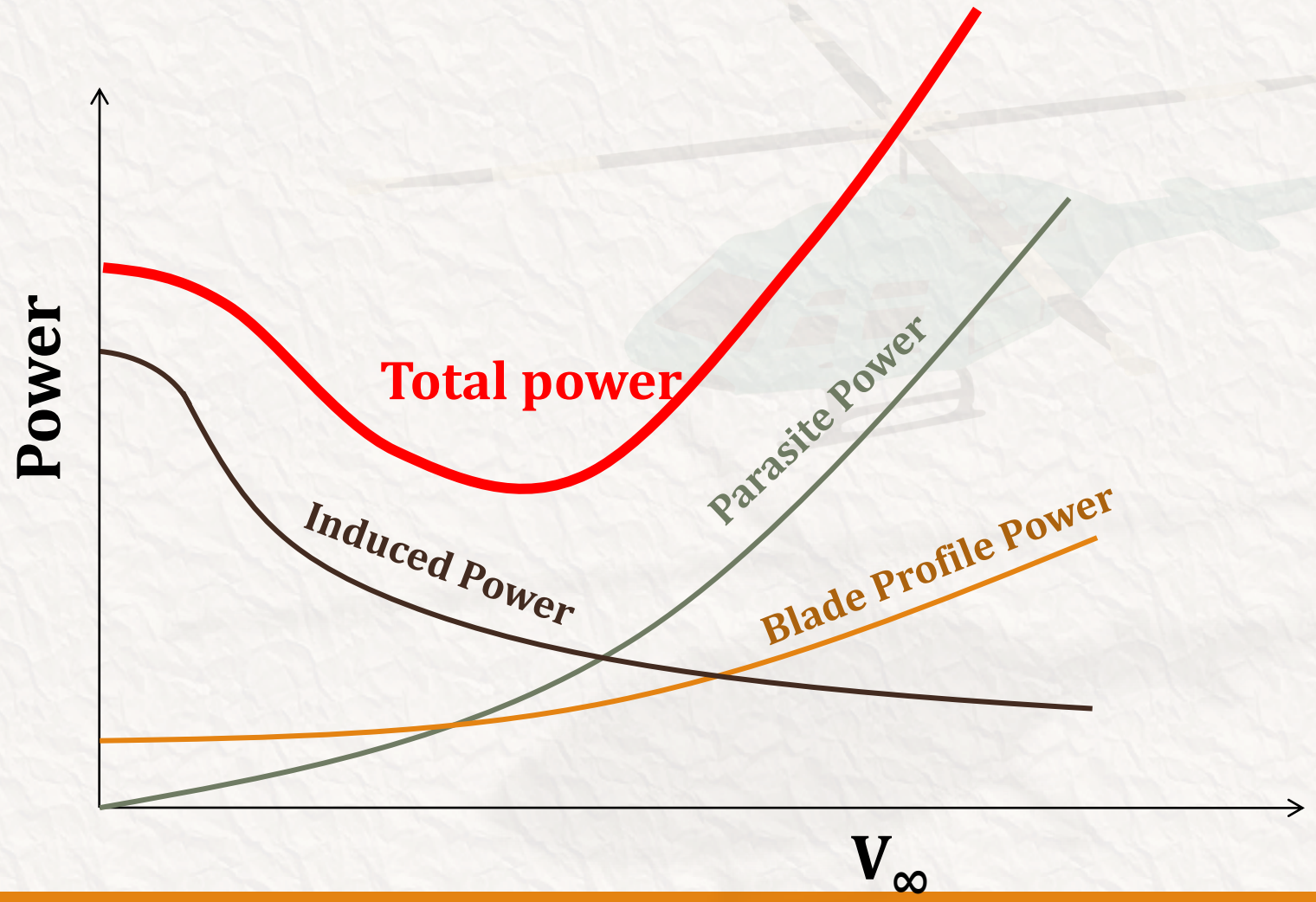
Power Estimate w.r.t Speed (Steady Level Flight)

- Total power = Induced power + Blade profile power
  $= DV_\infty + Tv +$ Profile Power
- Parasite power $(DV_\infty) = \frac{1}{2} \rho f V_\infty^3$
  (f is equivalent flat plate area)
- Induced Power $(Tv) \propto 1/\mu \propto V_\infty^{-1}$
- Blade Profile Power $= \rho A(\Omega R)^3 \frac{\sigma C_{d0}}{8}\left[1+3\mu^2\right] \propto V_\infty^2$

  C_{d0} is the average drag coeff. of airfoil

  (To be derived later)

Power Estimate w.r.t Speed (Steady Level Flight)



Power Estimate in Non-dimensional terms

$$P = P_{induced} + P_{parasite} + P_{profile}$$

$$\frac{P}{\rho A(\Omega R)^3} = \frac{T\mathrm{v}}{\rho A(\Omega R)^3} + \frac{\frac{1}{2}\rho V_\infty^3 f}{\rho A(\Omega R)^3} + \frac{\rho \mathrm{A}(\Omega \mathrm{R})^3 \frac{\sigma C_{d0}}{8}\left[1+3\mu^2\right]}{\rho A(\Omega R)^3}$$

$$\mathrm{C_P} = \frac{T}{\rho A(\Omega R)^2} \times \frac{\mathrm{v}}{\Omega \mathrm{R}} + \frac{1}{2}\frac{f}{A}\mu^3 + \frac{\sigma C_{d,0}}{8}\left(1+3\mu^2\right)$$

$$\mathrm{C_P} = C_T \lambda_i + \frac{1}{2}\frac{f}{A}\mu^3 + \frac{\sigma C_{d,0}}{8}\left(1+3\mu^2\right)$$

$$C_P = C_{P,i} + C_{P,parasite} + C_{P,0}$$

Empirical Corrections to C_P

Prediction through momentum theory does not account for non-uniform inflow, swirl, tip losses. Hence the expression is empirically corrected to match reality

$$C_P = C_{P,i} + C_{P,parasite} + C_{P,0}$$

$$C_P = C_T \lambda_i + \frac{1}{2}\frac{f}{A}\mu^3 + \frac{\sigma C_{d,0}}{8}\left(1+3\mu^2\right)$$

After empirical corrections :

$$C_P = \kappa C_T \lambda_i + \frac{1}{2}\frac{f}{A}\mu^3 + \frac{\sigma C_{d0}}{8}\left[1+4.6\mu^2\right]$$

Excess Power

- Power left after what is needed for steady level flight
- Can be used for climbing and accelerating
- Max achievable speed from energy perspective is when total power = available power
- Dependency of V_{Max} on air density

